

Renwei Meng

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EDUCATION

Anhui University, Hefei, China

Sep. 2023 – Jun. 2027

B.Eng. in Digital Media Technology

GPA: **3.95/4.0**; 4.34/5.0; 92.58/100; **Rank: 1/100**; **IELTS: 7.0**

Core Coursework: Diffusion & Flow Matching, Reinforcement Learning, Machine Learning, Linear Algebra, Probability Theory, Data Structures, Computer Organization, Operating Systems, Computer Networks.

RESEARCH INTERESTS

Trustworthy Generative AI (manifold-aware evaluation, reward hacking, reinforcement learning, flow matching); **Reliable LLMs and Agentic AI** (knowledge boundaries, abstention, RAG, multi-agent auditing); **Multimodal and Neurocognitive Intelligence** (EEG decoding, BCI, gaze modeling, synthetic-media detection).

PUBLICATIONS & PATENTS

Publications

* Equal contribution

- **Normal Fréchet Drift: A Geometric Measure for Detecting Off-Manifold Deviation in Generative Models.** Renwei Meng*, Yansen Han*, Tao Lin. *ICLR 2027*. **Under Review.**
Introduces a geometry-aware metric for quantifying harmful normal drift from pretrained data manifolds.
- **MORPH: Bi-Temporal Online Reinforcement Learning for Structure-Aware Music Flow Matching.** Renwei Meng. *AAAI 2027*. **Under Review.**
Performs counterfactual credit assignment jointly over flow steps and musical sections for long-form generation.
- **NeuroLens: Dual-Brain EEG for Past-Only Prefix Warning of Human-Recognized LLM Hallucinations.** Renwei Meng, Jingming Wang, et al. *AAAI 2027*. **Under Review.**
Combines inter-brain coupling graphs and causal Mamba decoding for EEG-based online warning of LLM hallucinations.
- **Know2Guess: A Contamination-Aware Multi-Zone Benchmark for Knowledge-Boundary Evaluation in Large Language Models.** Renwei Meng, Bowen Zhang, Jian Wang, et al. *ICONIP 2026*. **Accepted.**
Builds a 1,200-item, four-zone benchmark for answerability, abstention, refusal, and contamination.
- **Conditional Riemannian Adaptive Flow Transport for Cross-Subject EEG Decoding.** Renwei Meng, Hongyi Zhang. *PRICAI 2026*. **Accepted.**
Combines SPD alignment, neural residual flows, and label-free test-time adaptation across five benchmarks.
- **Group Resonance Network (GRN) for Cross-Subject EEG Emotion Recognition.** Renwei Meng. *ICANN 2026*. **Accepted.**
Integrates learnable group prototypes with PLV/coherence-based multi-subject resonance modeling.
- **S-AODP++: Structure-Aware AOI-Guided Scanpath Attention for AI-Generated Image Identification.** Jingming Wang*, Renwei Meng*, Yunjie Si*. *ICMI 2026*. **Accepted.**
Uses AOI transition graphs and scanpath attention, achieving 81.7% accuracy and 85.0% AUROC.
- **Explainable Innovation Engine: Dual-Tree Agent-RAG with Methods-as-Nodes and Verifiable Write-Back.** Renwei Meng. *arXiv preprint*.
Organizes methods in provenance and abstraction trees for controllable synthesis and verified write-back.
- **Breaking the Support Barrier in RLVR: A Fork-Conditional Reachability Criterion for Reliable Boundary Expansion.** Renwei Meng. *ACML 2026*. **Under Review.**
Introduces FCR-PO for predictable, verifier-gated support expansion at substantially lower compute than tree search.
- **VulnAgent-R2: Evidence-Calibrated Multi-Agent Auditing for Repository-Level Vulnerability Detection.** Renwei Meng*, Haoyi Wu*, Jingming Wang*. *ACML 2026*. **Under Review.**
Integrates risk screening, repository context expansion, specialist agents, and evidence fusion.

Patents

- **A generative deepfake video detection method and its system.** Shiang Hu, Zhiwen Zha, Renwei Meng, Yuhao Fang. China Invention Patent No. 202610165744.1. **Accepted.**
- **Resting-state EEG quality assessment based on dual-branch contrastive learning.** Shiang Hu, Dongdong Jia, Wan'er Lv, Renwei Meng, Chao Lv. China Invention Patent No. 202512044739.3. **Accepted.**

RESEARCH EXPERIENCE

Manifold-Aware Detection of Reward Hacking in Generative Models

Mar. 2026 – Present

Core Researcher / Advisor: Prof. Tao Lin / Westlake University

- Proposed **Normal Fréchet Drift**, which reconstructs local data geometry and separates tangential variation from harmful off-manifold drift, with theoretical guarantees and oracle-free hyperparameter selection.
- Validated NFD on analytical manifolds and text-to-image checkpoints; it detected degradation missed by reward, FID, nearest-neighbor, and divergence-based baselines. Manuscript under review at **ICLR 2027**.

Inter-Brain Synchrony for AI-Generated Content Detection		Sep. 2025 – Sep. 2026
<i>Core Researcher / Advisor: Prof. Shiang Hu</i>		
- Investigated AI-generated media detection through stimulus-locked cross-subject neural responses using dual-brain EEG hyperscanning experiments. - Built a neural-cognition-inspired framework using multi-band features, Transformers, GNNs, group prototypes, synchrony features, and Riemannian alignment. - Achieved 93.7% accuracy and 92.9% F1-score, outperforming CNN, SVM, and standard multimodal baselines.		
Industrial Metaverse Intelligent Interaction System		Sep. 2025 – Jun. 2026
<i>Algorithm Engineer / Advisor: Prof. Shiang Hu</i>		
- Built a real-time multimodal interaction system integrating eye tracking, gesture recognition, and pose estimation. - Designed low-latency intent-fusion algorithms and optimized 3D rendering and data transmission for industrial scenarios.		
Transformer-Based Software Vulnerability Detection		Mar. 2025 – Jan. 2026
<i>Project Leader / National College Student Innovation Program / Anhui University</i>		
- Developed Transformer-based vulnerability detection over AST and CFG representations to capture long-range code dependencies. - Used self-attention to extract deep semantic features from program structure and long-range execution context. - Curated a large-scale C/C++ dataset and evaluated precision and recall against static-analysis tools, leading to VulnAgent-R2.		
Selected Earlier Projects		
Coronary CTA Artifact Removal		Jan. 2025 – Jun. 2025
<i>Research Assistant / Advisor: Prof. Zhe Jin / Anhui University</i>		
Combined U-Net and GAN-based denoising to remove motion artifacts, improving SNR and SSIM over traditional filters.		
Carbon Price Prediction		Jun. 2024 – Jun. 2025
<i>Project Leader / National College Student Innovation Program / Anhui University</i>		
Led development of a CNN-GRU-LSTM-Attention model that improved prediction accuracy across seven Chinese carbon markets.		
INTERNSHIP EXPERIENCE		
Taiyuan Longwei Electronic Technology Co., Ltd. — Java Backend Developer Intern		Jul. 2025 – Sep. 2025
- Built Spring Boot/MyBatis services and RESTful APIs; improved data transmission efficiency and backend stability through frontend-backend collaboration. - Managed database operations and coordinated interface design with the frontend team for stable service integration.		
State Grid Shanxi Electric Power Co., Ltd. — Systems Operations Intern		Jul. 2024 – Sep. 2024
- Maintained critical power-system services and analyzed operational metrics for monthly optimization and reliability reports. - Supported platform availability and security for critical grid-management workflows.		
HONORS & AWARDS		
National Second Prize, National College Students Market Survey and Analysis Competition Finalist, Mathematical Contest in Modeling (MCM/ICM) National Scholarship (Top 0.1%, highest undergraduate honor in China) First Prize, National College Student Mathematical Modeling Competition (Provincial) Second Prize, Lanqiao Cup National Software Talent Competition (Java, Provincial) Honorable Mention, Mathematical Contest in Modeling (MCM/ICM) University Third-Class Academic Scholarship		2026 2026 2025 2025 2025 2025 2024
TECHNICAL SKILLS		
Programming: Python, MATLAB, C/C++, Java, R, SQL, HTML, Bash, zsh.		
Frameworks & Tools: PyTorch, TensorFlow, Git, Docker, Linux, E-Prime 3, LangChain, LangGraph.		
Additional: L ^A T _E X, PowerPoint, Figma, Visio, Adobe Photoshop/Illustrator, Blender (3D modeling); ChatGPT, Codex, Claude Code, n8n, ComfyUI.		